

Zemax

Title

ZPL_Code_Sequential_Jones_matrix_at_each_surface

Author

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Brief Description

This macro calculate Jones matrix at each surface for a single ray from specified (hx,hy,px,py).
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Overview

This macro traces a single ray through the system. The Jones matrix is calculated at each surface. The Jones matrix is based on a plan normal to the traced ray. The electric field value after each surface can be calculated by multiplying the electric field before the surface by the Jones matrix. In the macro, the result from Polarization Ray Trace (POLTRACE) is also shown for comparison.

Plugin Type

ZPL Analysis

Language

ZPL

Updates

Date	Version	OpticStudio version	Comment
18/10/2019	1.0	19.4SP2	Creation

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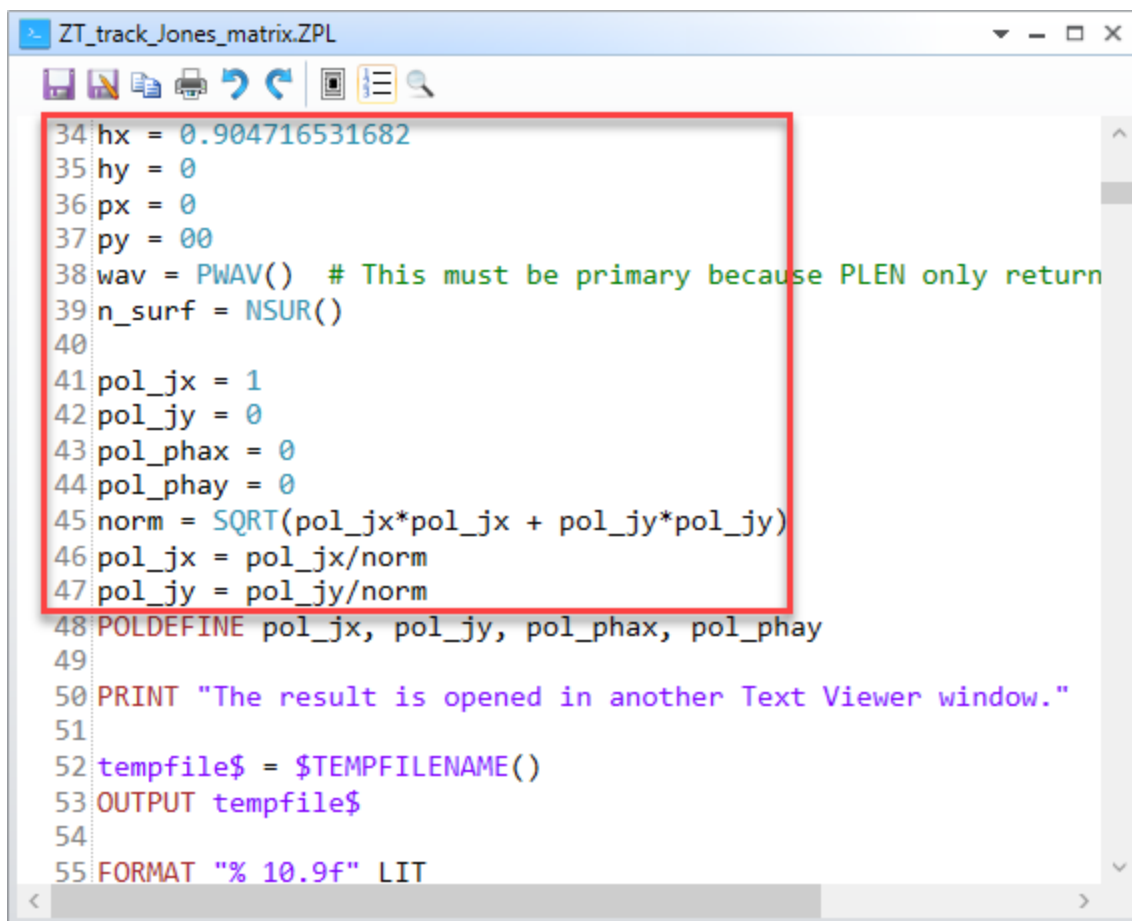
Instructions

1. Installation

Copy the macro to \Zemax\Macros\

2. How to use

Open the macro. Change the (hx, hy, px, py) and (jx, jy, phase_x, phase_y) as desired. And perform the macro.



```
34 hx = 0.904716531682
35 hy = 0
36 px = 0
37 py = 00
38 wav = PWAV() # This must be primary because PLEN only return
39 n_surf = NSUR()
40
41 pol_jx = 1
42 pol_jy = 0
43 pol_phax = 0
44 pol_phay = 0
45 norm = SQRT(pol_jx*pol_jx + pol_jy*pol_jy)
46 pol_jx = pol_jx/norm
47 pol_jy = pol_jy/norm
48 POLDEFINE pol_jx, pol_jy, pol_phax, pol_phay
49
50 PRINT "The result is opened in another Text Viewer window."
51
52 tempfile$ = $TEMPFILENAME()
53 OUTPUT tempfile$
54
55 FORMAT "% 10.9f" LIT
```

The result is like below:

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```
3: Text Viewer 3
Settings
Hx = 0.904716532
Hy = 0.000000000
Px = 0.000000000
Py = 0.000000000
Wave # = 1

jx = 1.000000000
jy = 0.000000000
phase x = 0.000000000
phase y = 0.000000000

===== Surface 0 (STANDARD) =====
Material: AIR

Rotation matrix (transition):
0.999991573    0.000000000    0.004105462
0.000000000    1.000000000    0.000000000
-0.004105462    0.000000000    0.999991573

veck = 0.004105462    0.000000000    0.999991573
vecjx = 0.999991573    0.000000000    -0.004105462
vecjy = 0.000000000    1.000000000    0.000000000
vecs = 0.999991573    0.000000000    -0.004105462
vecp = 0.000000000    1.000000000    0.000000000

jx = 1.000000000 + 0.000000000i
jy = 0.000000000 + 0.000000000i

angle (vecjx -> vecs) = 0.000000000

Jones matrix:
1.000000000 + 0.000000000i    0.000000000 + 0.000000000i
0.000000000 + 0.000000000i    1.000000000 + 0.000000000i

Accumulated Jones matrix at surface 0:
1.000000000 + 0.000000000i    0.000000000 + 0.000000000i
0.000000000 + 0.000000000i    1.000000000 + 0.000000000i

E-field (by Jones matrix, Global, GCRS = 1)
Ex: 0.999991573 + 0.000000000i    |Ex|^2 = 0.999983145
Ey: 0.000000000 + 0.000000000i    |Ey|^2 = 0.000000000
Ez: -0.004105462 + 0.000000000i    |Ez|^2 = 0.000016855

===== Surface 1 (STANDARD) =====
Material: AIR

Rotation matrix (transition):
```